

Blockchain Meets Biometrics: The Case of World(coin) from EU Data Protection Perspective and the Digital Omnibus

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Abstract. What if everyone on the planet were given an equal, unique share of a new cryptocurrency, with their iris as the proof of entitlement? The World Foundation, (previously “Worldcoin”) was set up by Sam Altman to become the most extensive global digital identity and financial network, anchored in biometric proof-of-personhood. Its operators scan irises and faces, derive an iris code, and issue a “World ID” together with cryptocurrency tokens. The project is pitched as a remedy for the surge of online AI-generated content, bots and fake accounts, a flood driven largely by the very generative AI tools the same actors are building. Multiple data protection authorities (DPAs) across the EU have conducted investigations and declared World(coin) data processing operations unlawful. We argue that the legal and ethical risks of World(coin) reach further than what the DPAs have so far articulated, and that the picture is shifting again with the European Commission’s Digital Omnibus package, which proposes a new Article 9 GDPR derogation for biometric verification. We start by looking at the objective and function of World together with EU enforcement record, and then analyse how the Omnibus proposals would change the legal analysis of World(coin) and similar deployments.

Keywords: Worldcoin, Biometric Data, Proof of Personhood, GDPR Digital Omnibus, Privacy-enhancing technologies, Identity Management.

1 Introduction

World Network or World, formerly Worldcoin, is a privately operated proof-of-human(Poh) network backed by Open AI’s Sam Altman. The website defines it as “*the real human network, an open-source protocol built to ensure every human benefits from the age of AI*”. Its premise is that online services can no longer reliably distinguish people from bots, AI agents, and synthetic identities, resulting in financial losses due to deepfakes and that by relying on iris data, reusable credentials attesting to human uniqueness can solve that problem. The World network combines a proof-of-human protocol, also known as proof of personhood (PoP), with a payments application (World App), the credential (World ID), and the application through which the credential is held and presented.[1]

Its operators scan irises and faces, derive an iris code, and issue a “World ID” together with cryptocurrency tokens. Biometric iris enrolment runs through the Orb, a purpose-built device that captures the face and eyes. During each new enrolment, the new template is compared against the templates of those already enrolled, not against a reference the user has supplied. World reports that more than 18 million people are registered via Orbs and World ID credentials issued while used more than 450 million times.[1]

World has promoted the credential for dating, gaming and event ticketing, where access may depend on one-person-one-account rules, and has entered integrations in the recent months including age assurance with Tinder and facial verification for enterprise communications with Zoom. Further proposed uses concern electronic signatures and confirmation that a human authorised an action taken by an AI agent.

The lawfulness of the underlying processing of iris data has been contested since its introduction. Supervisory authorities and courts inside and outside the European Union(EU) have ordered suspensions, corrective measures and fines, on grounds including the validity of consent, the enrolment of minors, transparency and the exercise of data-subject rights [CITE: BayLDA (under appeal); AEPD; CNPD; PIPC; ODPC Kenya; PCPD]. Most of those grounds are independent of the system's architecture. A distinct line of objection concerns the architecture itself: what is retained, which entities can reach it, and whether iris-derived templates remain personal data once technically transformed and distributed via PETs.

World has since changed that architecture. Enrolment data is held on the user's device. Comparison of protected representations is distributed across independently operated parties using secure multi-party computation. Zero-knowledge proofs allow attributes of the credential to be presented without disclosing the underlying representation to the relying party [CITE: World technical documentation; the identities and roles of the participating parties are documented separately]. These measures change where the information sits and which entity can reach it. They do not settle its legal status. Where processing is distributed in this way, classification cannot be resolved by asking whether one central actor holds a complete database.

The question is not confined to World. Reliance on biometric information in identity management is increasing across the EU. The revised eIDAS framework provides for European Digital Identity Wallets. The Commission has proposed digital travel credentials derived from passports and identity documents [CITE]. These systems differ from the World in purpose, legal basis and institutional setting while showing the direction towards more digitalisation in identity in several settings.

The applicable rules are themselves being reopened. Under the simplification agenda, the Digital Omnibus Package proposed amendments to the concept of personal data following the Court of Justice's judgment in *EDPS v SRB*, a mechanism for specifying when pseudonymised information ceases to be personal data, and a new Article 9 derogation for biometric verification where the biometric data, or the means required for verification, remain under the data subject's sole control [CITE: judgment with paragraph numbers; Commission proposal; Council compromise text, which removed the amendment to Article 4(1)]. Negotiations continue. In parallel, the European Data Protection Board's draft Guidelines 02/2026 on Anonymisation address how identifiability

is to be assessed by reference to a particular entity and a particular processing operation [CITE].

This article uses World as a stress test of the proposed reform in the EU GDPR and the recent guidelines. It asks how the Digital Omnibus and the developing interpretation of anonymisation would change the classification of the biometric information processed within World ID, and whether and how the proposed biometric verification derogation under Article 9 GDPR would affect the ongoing World systems.

Section 2 describes World's successive architecture following the DPA decisions by the BayLDA and outside the EU. Section 3 examines the World's data processing operations in the light of EU case law on personal data and under the proposed amendments in the DOP. Section 4 ends with concluding remarks.

2 “In progress” from Worldcoin to New World: Regulatory Pressure and Architectural Redesign

The Orb performs the initial enrolment as a first step to get the World ID. It captures images of the user's face and eyes and derives a numerical representation of the iris, referred to as an iris code. The system then compares the new iris code against the enrolled population. The purpose is to determine whether the person has already registered, and this process is called "duplicate biometric enrolment check" or more generally "de-duplication checks".[2] Since there is no reference template previously supplied by that user, the new representation must therefore be compared against the representations of existing enrollees. This uniqueness check differs from later biometric authentication, where an individual uses World ID to access services. Later authentication can instead compare the current user against a reference associated with the verified World ID. The legal classification of these operations biometric identification(1:n) vs verification (1:1) is addressed in Section 3.

World's earlier architecture relied on a centrally managed database of iris codes to support the so-called population-level deduplication checks but World notes that access was restricted through encryption, hardware security modules and organisational controls. In March 2024, the Spanish and Portuguese supervisory authorities temporarily restricted World's biometric processing.[3] Authorities in Hong Kong and South Korea later adopted enforcement measures.[4] The BayLDA completed the first stage of its EU investigation in December 2024. Its decision required changes concerning erasure and consent, while other issues remained under examination or appeal. These proceedings addressed more than the security of stored iris codes. The published decisions and notices also concerned transparency, consent, children's data and the exercise of data-subject rights such as right to erasure.

Following the DPA decisions, World introduced Personal Custody in March 2024. Under the published design, the Orb encrypts facial and iris images, metadata and derived information generated during enrolment. The encrypted package is transferred to the user's device and the Orb deletes its local copy after the transfer. Hence, Personal Custody, which attempt to allow users to be in control of their biometric data, changed where enrolled biometric data were retained. However, this did not remove the need to

compare a new iris code against those already enrolled. World therefore introduced secure multi-party computation (SMPC) in 2024 for the deduplication stage.

Under SMPC, each iris code is divided into secret shares. Separate computing parties receive different shares. They perform the comparison jointly without first reconstructing the complete iris codes in one location. World announced in May 2024 that it had deleted the former central database consisting of iris templates after deploying this system. The first deployed SMPC arrangement involved entities connected to Tools for Humanity and the World Foundation. World later replaced it with what it calls anonymised multi-party computation (AMPC) and the migration was completed in January 2025. World argues that AMPC retains secret sharing but changes the output of the comparison and reveals only whether a match was found. The iris codes and associated masks remain divided between the nodes during the calculation. The participating organisations also changed that neither Tools for Humanity nor the World Foundation operates an AMPC node. The AMPC node operators consist of commercial entities and university research centres, including the University of Erlangen-Nuremberg (FAU), UC Berkeley Centre for Responsible Decentralised Intelligence (RDI), the Korea Advanced Institute of Science and Technology (KAIST), the University of Tokyo and Nethermind, including commercial entities and university research centres. World also states that the list may change over time. The protocol and parts of its implementation are all publicly available. The privacy properties nevertheless depend on organisational assumptions. Each participant must hold only its assigned share and follow the agreed protocol. The protection against reconstruction depends on the required number of parties not combining their information outside that protocol.

World relies on these features to describe the AMPC shares as anonymous. Its argument is that an individual node receives neither the complete iris code nor sufficient information to attribute its share to a person and the nodes receive only the result required for deduplication. Overall and to the extent relevant to this paper, the New World re-design changed the flow of iris data at two points. Enrolment data moved to the user's device. Complete iris codes were replaced by distributed shares for deduplication checks.

No published supervisory decision has yet assessed the current AMPC arrangement in its deployed form. The BayLDA decision concerned an earlier architecture and the changes made during its investigation. The following section will analyze the impact of using such PETs.

3 World's data processing operations in light of EU case law on personal data and under the proposed amendments in the DOP

3.1 Argument 1

World argues that their iris processing approach, decentralized storage solutions and SMPC/AMPC application answer most personal data challenges under the EU GDPR by ensuring efficient anonymisation. The Bavarian Data Protection Authority

(BayLDA) decided in December 2024 that this was not the case and the iris codes and secret shared encrypted under SMPC were personal data. Importantly, the changes proposed under the DOP might change the current interpretation regarding controllership and personal data and therefore are relevant for our assessment. As will be explained below, we believe it is still possible to argue that the data processed by World is personal data as the parties might have ‘the means’ that are reasonably likely to be used to identify the data subjects.

To begin with, it should be assessed whether the World’s proposed decentralised storage solutions and SMPC application would allow in light of the DOP to categorise the processed data as anonymous. If it would be deemed still personal data, for which actors would this be the case? Would the contractual arrangements between World entities and further division of processing to universities achieve non-identifiability? We argue that the key question to address here is whether the processed biometric image and the template qualify as personal data considering the recent legal developments.

Gdpr - the definition of personal data. The GDPR takes a comparatively objective (absolute) approach in defining personal data.[5] Personal data is defined as any information relating to an identified or identifiable natural person. Recital 26 clarifies that in assessing whether a person is identifiable, one must consider all means reasonably likely to be used whether by the data controller or by any other party, to identify them either directly or indirectly.

World argues that, in theory and technically, encrypted iris code (secret shares) is not reconstructable due to this security measure in the form of the SMPC. We will briefly analyse this argument from the perspective of both the absolute approach and the relative approach to identifiability, in light of the recent case law and the newly published European Data Protection Board (EDPB) guidelines on anonymisation.[6]

The absolute or the “simplified” approach. The absolute approach (or “the simplified approach” according to the new EDPB guidelines on anonymisation[6]), considers data identifiable when any party, regardless of the controller, can identify the data subject. Accordingly, as long as there is a possibility for the re-identification, the data at hand is deemed personal. As for SMCP, this method does not completely preclude the possibility of reconstructing the personal data at the origin. For instance, if an attacker (or a legal order) can compel decryption keys from several participating parties at once, they may be able to decrypt the secret shares and reconstruct personal data. Therefore, according to the absolute approach, SMCP is merely a decentralizing security measure and not an anonymization technique.

The relative or the “contextual” approach. The relative approach or “the contextual approach” as named by the EDPB[6], takes a narrower stance and considers specifically the means available to the entity who processes the data, by taking into account the specific facts of the situation. Below, we will analyse the World’s processing operations within this approach.

Some seminal CJEU decisions are worth mentioning here for our analysis in relation to the relative approach. To start with, in Breyer,[7] the Court held that, in the context of dynamic IP addresses, information can qualify as personal data even where the data necessary to identify the individual is not all held by one single entity (para 43). To assess whether a person is identifiable, all means reasonably likely to be used must be

taken into account (para 42). In this case, the entity in question had legal means to obtain additional data from a third party that could enable identification. The Court held that data qualifies as personal even where identifying an individual would require recourse to legal means. This extends to scenarios where a data controller might approach a competent authority to obtain data through legal proceedings, in order to identify relevant individuals, for instance, in the aftermath or during a cyberattack (para 47).

In *Scania*,^[8] the Court held that a vehicle identification number (VIN) constitutes personal data where a party (repairer) possesses the means to link it to a specific natural person. While a VIN may not, in and of itself, constitute personal data for every entity that holds it, it can nonetheless acquire that status indirectly once it is disclosed to another party equipped with the means to identify the individual concerned. (para 49).

Finally, in *SRB*,^[9] the Court ruled on pseudonymized data, which is relevant to the subject of this article as encrypted data is usually considered pseudonymized.^[10, 11, 12] The case concerned whether pseudonymized shareholder comments that were gathered in connection with a bank resolution process qualified as personal data after it is shared with a third-party auditor. The Court found that personal opinions expressed in the comments are inherently linked to individuals and the SRB retained the means to identify the commentators, hence, the information in question constituted personal data from the SRB's perspective.

The relative approach emphasizes the necessity of considering each situation in its own context, taking into account all the means that are available to controllers or processors, and whether these means are reasonably likely to be used. In the case of *World*, the analysis concerning the identifiability of the data processed by *World*, according to the relative approach, is not straightforward due to so-called decentralized features (in particular of SMPC) of the technology. SMPC operates with secret shares that are available to different parties. These are encrypted pieces of the personal data at the outset and are not identifiable, the additional identifying data might be obtained from a third party (or transmitted to a recipient). If the parties who are supposed to be independent instead cooperate covertly, they can regenerate the secret without the user's knowledge or consent.^[13] If different secret shares can be brought together through cooperation or collusion between different parties, the data may become personal again (see *BayLDA* para 345). The key question is who holds the pieces and whether they can be made to cooperate. In *World*'s SMPC infrastructure, a couple of parties involved in Shamir's secret sharing, including *Tools for Humanity* and the *Worldcoin Foundation's* EU entity, are closely affiliated with each other rather than being fully independent parties (see *BayLDA* para 346-348). It is crucial to note that these parties already agree to work together, that they are parties to a business. The technical infrastructure does not protect against the potential reconstruction through their collaboration, therefore it does not preclude identifiability. In this technical context, only institutional independence and legal barriers to compelled disclosure do.

By the same vein, *Bavaria's BayLDA* stated that *World's* processing of iris images entails a number of fundamental data protection risks for a number of data subjects, and it does not comply with GDPR.^[14] Accordingly, an order to erase presupposes something capable of being erased because it remains linked to a person, not something already rendered anonymous.

However, under the DOP amendment of personal data this legal claim can be more likely to be true considering DOP proposes to amend the definition of personal data by adding the following paragraph: *“Information relating to a natural person is not necessarily personal data for every other person or entity, merely because another entity can identify that natural person. Information shall not be personal for a given entity where that entity cannot identify the natural person to whom the information relates, taking into account the means reasonably likely to be used by that entity. Such information does not become personal for that entity merely because a potential subsequent recipient has means reasonably likely to be used to identify the natural person to whom the information relates.”* (emphasis added by author, DOP Art 3(1)(a)).

3.2 Argument 2: World's biometric verification would still fall under the GDPR Article 9(2)

With regard to the initial biometric data processing, the Orb processes iris images only locally and deletes them after extracting the code. No raw biometric image is stored. This processing still falls under the definition of biometric data under the GDPR, as the GDPR applies exactly to the biometric references and templates that went through specific processing rather than biometric images in Article (14).

The current version of the GDPR applies to both verification and identification functions, and these are listed among the prohibited processing operations in Article 9.

The World employs a bundle of decentralized solutions that can nuance our answer. Instead of one company running a centralized 1:many match against a database under its full control, the uniqueness checks are split across parties who each see encrypted fragments of the biometric template. The comparison is done too on these encrypted fragments. This itself does not change the fact that the uniqueness check is being done. Moreover, the Orb is still connected to a centralized hardware and enrolment system that is later transmitting biometric data into a cryptographically split multi-party storage and computation layer, which can be seen as a security measure.

The DOP introduces a new exemption under Article 9(2) GDPR to allow processing of biometric data where necessary to confirm the identity of a data subject (verification) or to authenticate individual, provided that biometric data remain under the sole control of the data subject.¹ This exemption brings two main issues to our attention: 1) The distinction between identification and verification; 2) the condition for the biometric data to remain under the “sole control”.

Verification vs identification. The proposed provision draws an explicit distinction between verification and identification. In other words, it aims to create an additional legal basis for processing biometric data particularly for biometrics verification that aims to establish the claimed identity of individuals.

¹ The text of the proposed amendment is as follows: “(1) processing of biometric data is necessary for the purpose of confirming the identity of a data subject (verification), where the biometric data or the means needed for the verification is under the sole control of the data subject.”

Before March 2024, the World's architecture was not fully decentralized; it had centralized and distributed features. The biometric template was not sitting in the user's own device under their sole control. From March 2024 on, the Foundation allegedly changed this to a more decentralized system where user control is provided by storing the iris code in user's mobile devices.[15] This change would not change the status of the World's biometric data processing according to the current version of the GDPR Article 9. Even if considered as verification, such processing would still be subject to the prohibition in Article 9 special categories of data. However, if the DOP provision passes as drafted, it is possible for the Foundation to argue that special category prohibition in Article 9 would not be applicable, since biometric data are stored only in the user's device.

On the other hand, as we explained above, the deduplication checks are still done with 1:many comparison to ensure the user is not registered to the system before, which would still fall under Article 9 restrictions, even with the DOP exemption. It is important to note that the World Foundation describes deduplication checks as a necessary step to avoid fraud and ensure trust towards the identity by comparing the new Iris codes against the ones stored before, currently consisting of more than 18 million samples. This raises the question of whether one of the already existing exemptions under Article 9 could apply to such processing. Currently, there is no guidance on how this very common procedure will be categorized under the GDPR. In our view, due to the highly sensitive nature of biometric data and the risks related to their processing, such processing for biometric verification purposes should still be subject to the general prohibition of Article 9. Moreover, it should not be neglected that biometric verification systems operate with unique identifiers that are often connected to user accounts that consist of various types of personal data depending on the service provided. In this case, third parties indeed use such unique identifiers relying on the verification carried out through World ID. While these unique identifiers can both be special categories or regular personal data, their processing is still subject to the GDPR and requires a valid legal basis. Identifiability of this type, i.e., biometric verification connected to user account data vaults, is not risk-free. First of all, biometric data function as the key to the rest of the digital identity of the data subjects. Compromise of these data would result in the leakage of the rest of the data vault, often without available means to become aware of it or contest by the data subject. Where BV is used, it provides certainty that the person is indeed the one originally registered, for example, through a uniquely assigned identifier.

Second, biometric verification systems might not eliminate surveillance risks that are often attributed to biometric identification systems. As Renieris writes, 'even as the digital realm becomes more invisible, human beings are becoming more machine readable through the proliferation of biometric-enabled digital identity tools and technologies.' Biometric verification systems still require capturing and storing biometric data, which can later be repurposed, linked across contexts, or accessed by third parties. Moreover, the infrastructure, such as the sensors, templates, or algorithms, can be redirected towards identification functions. Thus, it is plausible to argue that the distinction between these two functions may be more fragile in practice. Consequently, it should

not be neglected that framing biometric verification systems as inherently privacy-preserving risks surveillance capacity they actually create.

4 Is there any hope for PETs to achieve meaningful anonymisation?

It is important to emphasize that the BayLDA does not argue that anonymisation cannot ever be achieved via Privacy enhancing technologies such as SMPC. It rather states that the specific technologies and security measures deployed by World at the time of the decision are not sufficient to achieve anonymisation.

In light of the BayLDA's nuanced approach to PETs, the recent case law including the SRB case, and the anonymisation guidelines by the EDPB, it may indeed be possible to achieve anonymisation, especially from the perspective of the relative or the "contextual" approach. [Ezgi to add a brief summary of requirements from EDPB + ref's to EDPB's examples: the three criteria of No Record Isolation, No Linkage, No Inference, applied for the contextual approach].

Similar to the EDPB, the Norwegian Data Protection Authority has also acknowledged the future potential of privacy enhancing technologies such as fully homomorphic encryption (FHE), to increase the security of encryption and encrypted data, "as well as its potential use in other areas that can benefit from analysing and processing data while preserving confidentiality" and therefore without revealing personal data.²

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1 st -level heading	1 Introduction	12 point, bold
2 nd -level heading	2.1 Printing Area	10 point, bold
3 rd -level heading	Run-in Heading in Bold. Text follows	10 point, bold
4 th -level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], as well as a URL [5].

² Datalysnet, "Securing Digital Identities: Biometric data and protected templates in eID solutions - Exit report from the "SALT" sandbox project with Mobai", < <https://www.datatilsynet.no/contentassets/fad759b2486c41bf88dd58d8743a3a7a/salt-exit-report.pdf> > accessed 31 July 2026.

5 Conclusion

The legal proceedings and DPA decisions concerning World's data processing have demonstrated that data protection law serves as a foundation for public intervention against biometric digital identity systems managed by private entities. As highlighted above, authorities within and outside the EU have issued enforcement decisions, corrective orders and fines, while several proceedings remain pending, exposing serious problems in World's large-scale collection and use of iris data and facial images.

The regulatory backlash also pushed World to redesign its architecture and introduce new safeguards and PETs. Yet it is hard to argue that relying on PETs resolves all the legal issues resulting from World's biometric data processing operations, and not all of them were about the security of the biometric data at stake. Questions concerning consent, economic inducement, transparency and large-scale biometric enrolment remain open and while remain connected and hinge upon the applicability of the GDPR and the legal nature of the biometric data processed by World.

The Digital Omnibus now amends the instrument under which those interventions were made. This article has examined two of its elements: the treatment of the personal-data concept and the proposed derogation for biometric verification under Article 9.

On the first, the proposed approach to identifiability, if the Commission text prevails on this point, would have consequences across biometric and other identity systems built on privacy-enhancing technologies. It strengthens the position of a private actor asserting that distributed processing has rendered the information anonymous from its own perspective. Where that assertion succeeds, Article 4(14) cannot be reached and the Article 9 regime falls away with it. Ultimately, this will weaken the data subject rights and protections granted by the GDPR and enforced by supervisory authorities, which initially compelled World to redesign its systems in accordance with applicable data protection rules.

On the second, there is little doubt that Orb enrolment constitutes processing of personal data. The device collects biometric characteristics and compares them against an enrolled population to establish uniqueness. The open question concerns what happens to the iris codes afterwards, under the multi-party arrangement.

The protocol is published and its implementation is open source. The difficulty is therefore not one of disclosure. It is that what is disclosed is a set of organisational assumptions rather than a technical guarantee. Privacy holds for as long as the parties do not collude, and multi-party computation cannot itself enforce that condition. The parties are a mix of universities and commercial entities and one of them also stores the share database. World states that the participating institutions may change over time. No supervisory authority inside or outside the EU has yet assessed this arrangement.

Whether the shares are pseudonymised or meet the anonymisation criteria in the EDPB draft guidelines therefore turns on those arrangements as they stand. Two consequences follow if they are personal data. Nodes operated outside the EU raise a international data transfer question. And the parties run on a common cloud provider, which World identifies as AWS. The independence on which the anonymity via MPC argument rests is weaker than the number of parties suggests, since this increases the

attack surface correspondingly. The Bavarian authority raised comparable concerns about concentration of access.

Second, the newly proposed derogation for biometric verification under Article 9 GDPR could provide World with an alternative condition for processing special-category biometric data. It would not, however, replace the need for a legal basis under Article 6 GDPR. Its relevance lies in World's current reliance on explicit consent under Article 9(2)(a), which has been contested because token incentives may affect whether consent is freely given.

The derogation could facilitate later verification based on an existing World ID. It would not cure defects in the biometric enrolment through which that credential was created and the de-duplication check. This distinction matters where the initial collection of iris data was based on disputed consent. The strongest counterargument is that enrolment and subsequent verification are separate processing operations and may rely on different legal grounds, but this is still to be seen.

World's recent report published in June 2026, named *The Simple Plan*, states that the project entered the third phase, and among other priorities, taking the pragmatic approach, they will concentrate operations on a core set of countries where proof-of-human products already show early adoption, among them Germany and the United Kingdom. These jurisdictions have relatively robust data protection laws, in particular the EU GDPR, and DPAs have worked on the World's contested technology. The system is likely to be built out under close scrutiny at the same time as the data protection rules governing it in the EU are being renegotiated, and World continues to add more services and biometric modalities while that is unresolved. How the Digital Omnibus settles the personal-data definition and the biometric verification derogation will definitely shape what World operations will look like in the EU. But without doubt, World's development will remain an interesting regulatory stress test for the robustness of data protection laws and fundamental rights in the EU and outside.

Acknowledgments. A third level heading in 9-point font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Imec project

Mobai project

Bilgesu's project(?) scamper?

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References

1. Worldcoin website, https://worldcoin-company-website.cdn.prismic.io/worldcoin-company-website/Zuvl4bVsGrYSvIIU_WorldcoinPrivateByDesign.pdf, last accessed 2026/07/31

Field Code Changed

2. ISO/IEC 2382-37:2022(En), Information Technology - Vocabulary - Part 37: Biometrics' <<https://www.iso.org/obp/ui/en/#iso:std:iso-iec:2382:-37:ed-3:v1:en>> last accessed 2024/08/30 - **Duplicate biometric enrolment check 37.08.04**
3. 'Worldcoin pivots away from Europe amid tangle of GDPR problems | Biometric Update', <https://www.biometricupdate.com/202410/worldcoin-pivots-away-from-europe-amid-tangle-of-gdpr-problems>, last accessed 2026/07/31
4. 'Worldcoin rolls out in Guatemala, receives fine from Korea | Biometric Update', <https://www.biometricupdate.com/202409/worldcoin-rolls-out-in-guatemala-receives-fine-from-korea> last accessed 2026/07/31
5. Corte, LD: Scoping Personal Data: Towards a Nuanced Interpretation of the Material Scope of EU Data Protection Law. European Journal of Law and Technology 10 [sayfalar] (2019)
6. EDPB, 'Guidelines 02/2026 on Anonymisation – Version 1.0', https://www.edpb.europa.eu/system/files/2026-07/edpb_guidelines_202602_anonymisation_v1_en_0.pdf, last accessed 2026/07/31
7. [BREYER]
8. [SCANIA]
9. [SRB]
10. EDPB, 'Guidelines 01/2025 on Pseudonymisation', https://www.edpb.europa.eu/system/files/2025-01/edpb_guidelines_202501_pseudonymisation_en.pdf, last accessed 2025/11/19
11. ENISA, 'Pseudonymisation Techniques and Best Practices Recommendations on Shaping Technology According to Data Protection and Privacy Provisions' (2019) Recommendations;
12. AEPD, 'INTRODUCTION TO THE HASH FUNCTION AS A PERSONAL DATA PSEUDONYMISATION TECHNIQUE' (2019) Opinion
13. Silence Laboratories website, <https://silencelaboratories.com/blog/a-discussion-on-collusion-resistant-protocols-in-secret-sharing>, last accessed 2026/07/31
14. [empty ref]
- 15.
16. Author, F.: Article title. Journal 2(5), 99–110 (2016)
17. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) CONFERENCE 2016, LNCS, vol. 9999, pp. 1–13. Springer, Heidelberg (2016)
18. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
19. Author, F.: Contribution title. In: 9th International Proceedings on Proceedings, pp. 1–2. Publisher, Location (2010)
20. LNCS Homepage, <http://www.springer.com/lncs>, last accessed 2023/10/25